Bubble fluctuations

X18 (2018) — English translation

Consider an air bubble of radius R immersed in water of density ρ. At equilibrium, the pressure inside and outside the bubble is p_0, if the surface tension of water is not taken into account. Consider air to be an ideal diatomic gas. After a slight disturbance, the bubble begins to oscillate radially with a certain frequency. The following steps will help you find this frequency. This can be done by representing the potential energy of an oscillating bubble as Ax^2, and the kinetic energy as $B(dx/dt)^2$, where x is a small change in radius. Then the frequency will be $\omega = \sqrt{A/B}$.

A1	Find the restoring pressure force acting on a small area dS of the bubble surface when its radius changes adiabatically from R to $R+x$, ($x \ll R$).	**0.70**

A2	Find the energy required to change x from 0 to some small value x_0.	**1.00**

A3	The radial speed of water at the surface of the bubble is dx/dt. Given that water is incompressible, find the radial speed of water at a distance r ($r > R$) from the center of the bubble.	**1.00**

A4	Find the total kinetic energy of water.	**1.00**

A5	If we neglect the air mass, the kinetic energy found in the previous paragraph will be the kinetic energy of the oscillating bubble. Find the vibration frequency.	**0.50**

Now let's take into account the influence of surface tension. The surface tension coefficient σ is defined as the energy required to create a surface of unit area ($W = \sigma S$).

B1	Find the contribution of surface tension to the energy required to create a bubble of radius R in water.	**0.20**

B2	Find the frequency of bubble oscillations, taking into account the influence of surface tension of water. Consider the pressure outside the bubble equal to p_0.	**1.60**

Source: https://pho.rs/p/311